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Influence of chemical bonding on thermal contact resistance at silica interface: A molecular dynamics simulation

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ABSTRACT

The chemical bonding is usually not avoidable at a welding or fusion interface and sometimes is desirable to enhance the mechanical strength, while it might strongly affect the thermal contact resistance which plays a critical role in hindering heat dissipation of electronic devices. In this paper, we employ a non-equilibrium molecular dynamics method to study the effect of chemical bonding on the thermal contact resistance. Results show that the chemical bonding can greatly affect the thermal contact resistance. With the increase of the chemical bonding ratios, the thermal contact resistance firstly drops dramatically until arrives at a bonding ratio of about 20% where the thermal contact resistance is only about 15% that with a bonding ratio of 0%, and then gradually decreases to 0. Analyzing the spectral energy density of phonons at the interface, we conclude that the increase of the phonon velocity should be responsible for the decrease of the thermal contact resistance. By simulating the thermal contact resistance of silica with different morphologies (crystalline or amorphous silica), we found that different morphologies could lead to different thermal contact resistances, although this difference is less than 6%. This study is expected to provide references for managing thermal transport at a welding or fusion interface.

1. Introduction

The chemical bonding is usually not avoidable at a welding or fusion interface and sometimes is desirable to enhance the mechanical strength. To reinforce the structure strength, the chemical bonding has been widely established with a high pressure and/or a high temperature to bond metals, alloys, ceramics and composites [1–3], and the chemical bonding can also occur in a process of annealing [4]. Besides above methods, severe mechanical stress was also applied to build an interfacial bridging by Si–O–Si bonds between spherical Si nanocrystals and a α-SiO_2 matrix [5]. Although there are always chemical bonds at interfaces, the effect of chemical bonding on thermal contact resistance has been rarely investigated yet. More importantly, the chemical bonding can greatly affect the thermal contact resistance, while thermal contact resistance plays a critical role in hindering heat dissipation in electronic devices.

Numerous studies have already been performed to probe methods to reduce the thermal contact resistance [6–9]. Most recently, it is reported that the presence of chemical bonds could greatly decrease the thermal contact resistance between amorphous silica nanoparticles [10]. In this paper, the chemical bonding effect on thermal contact resistance is further studied. Considering the wide application of silica in thermal insulation [11–13] (silica aerogel or silica composites) and electronic packaging [14] (silica packing material), the thermal contact resistance between crystalline-crystalline, amorphous-crystalline, and amorphous-amorphous silica is focused in this paper. The α-SiO_2 is applied as the crystalline silica in this work.

The non-equilibrium molecular dynamics (NEMD) simulation is firstly applied in this work to get the temperature distribution along heat transfer direction with a given heat flux, where a temperature jump will happen at the interface because of the contact resistance [6,15]. Then, according to the definition, the thermal contact resistance is calculated by the temperature drop divided by the heat flux. Finally, analyzing the cumulative correlation factor (CCF) which depicts mismatch of vibrational density of states (DOSs) between two materials, and radial distribution function (RDF) which represents the difference of structures, the structure dependence of thermal contact resistance is explained. The spectral energy density (SED) of phonons at the interface is also calculated to illustrate the effect of chemical bonding on phonons. Results turn out that the chemical bonding can greatly affect the thermal contact resistance. This study is expected to provide references for managing thermal transport at a welding or fusion

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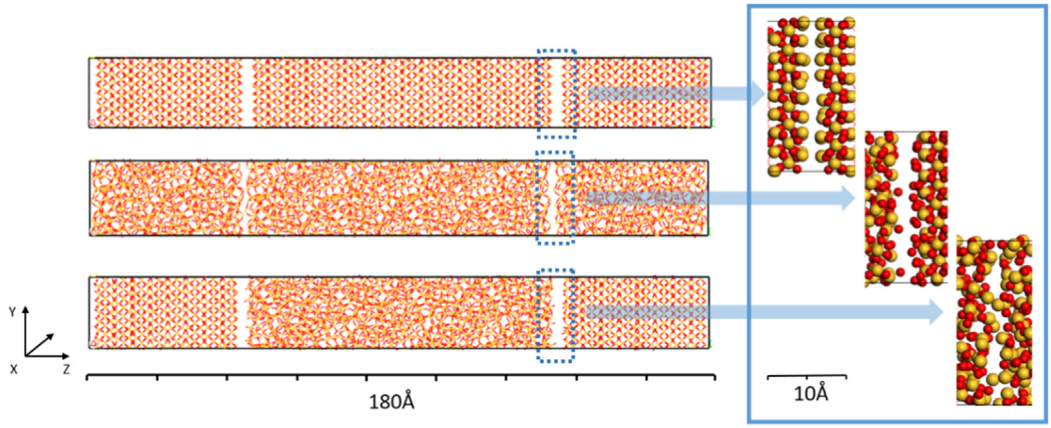

Fig. 1. Structures used in our simulation (interface without chemical bonds is shown here). The upper, middle and lower panels show crystalline-crystalline, amorphous-amorphous, and crystalline-amorphous interface configurations respectively.

interface. With a known thermal contact resistance, the revealed relationship between the thermal contact resistance and bonds can be also applied to estimate the interface structural strength.

2. Simulation and calculation methods

Sandwich structures are built to investigate the thermal contact resistance with Material Studio package, as shown in Fig. 1. Every structure contains about 5000 atoms. As that done in the previous work [16], an interface distance of about 2 Å is applied here. The structure size used in our simulation is summarized in Table 1. For comparison, three kinds of silica contact interface are studied, the amorphous-amorphous, crystalline-crystalline and crystalline-amorphous interfaces.

For convenience, the interfacial bonding ratio is defined as the percentage of atoms connected through Si—O—Si bonds to other-side atoms at interface. The calculation method of the interfacial bonding ratio is added in Appendix A. To illustrate the influence of chemical bonds on thermal contact resistance, different bonding ratios are applied, i.e., 0, 6.25, 12.5, 25, 50 and 75%. Chemical bonds are distributed randomly at interface in our simulation, as that done in Ref. [17]. A crystalline-crystalline configuration with a bonding ratio of 6.25% is shown in Fig. 2.

In the present work, the COMPASS potential [18–20] which is the first ab initio force field and has been validated to be capable of accurately predicting structural and thermophysical properties within a broad range of organic and inorganic substances, is used to describe the interactions between atoms. The potential function is exhibited in Appendix B. In general, the terms in the potential function could be divided into bonded interaction terms and non-bonded interaction terms. While there are no bonds at interface, two materials at interface are connected by van der Waals force and Coulomb force described by extended L-J equation,

$$\mathrm{V(r)} = \sum_{i,j} \varepsilon_{ij}\left[2\left(\frac{\sigma_{ij}}{r_{ij}}\right)^9 - 3\left(\frac{\sigma_{ij}}{r_{ij}}\right)^6\right] + \frac{1}{4\pi\varepsilon_0}\sum_{i,j}\frac{q_i q_j}{r_{ij}} \tag{1}$$

where r is the distance between two atoms located at different side of interface, q_i and q_j are the signed magnitudes of the charges, ε_0 is

Table 1
Structure size applied in the simulation.

Structure	X direction (Å)	Y direction (Å)	Z direction (Å)
Crystalline-crystalline	17.50	20.21	179.24
Amorphous-amorphous	20.95	20.95	171.41
Crystalline-amorphous	19.10	20.42	175.31

Coulomb's constant, $\sigma_{ij} = \left[\frac{\sigma_i^6 + \sigma_j^6}{2}\right]$ and $\varepsilon_{ij} = 2\sqrt{\varepsilon_i \varepsilon_j}\frac{\sigma_i^3 \sigma_j^3}{\sigma_i^6 + \sigma_j^6}$, where ε is energy parameter, σ is the distance parameter. The L-J parameters are taken from Ref. [18] listed in Table 2.

Before NEMD simulation, the configurations should be optimized to confirm the energy minimization. Geometry optimization is firstly performed on the initial configuration, and a NPT ensemble is applied to release the structure interstress for 50 ps with a time step of 0.5 fs, at a pressure of 0.1 MPa controlled by Berendsen method and a temperature of 300 K controlled by Nose-Hoover thermostat. Then, the systems are equilibrated for 500 ps with a time step of 0.2 fs at 300 K under the NVT ensemble. We double-check that the total energy has reached minimum and convergence has been reached at the end of NVT ensemble to make sure that our systems have already been equilibrated.

Finally, with a periodic boundary applied in the X, Y and Z directions, where Z direction is set to be the heat transfer direction, the NEMD method [21–25] is applied to get the temperature profile. In this work, with every 10 time steps, the velocity of the fastest atom in "cold" slab is replaced by the velocity of the slowest atom in "hot" slab, and vice versa. Consequently the "cold" slab becomes colder, whereas the temperature of the "hot" slab increases. The system will react by flowing energy from the hot to the cold region. Eventually a steady state sets in when the energy exchanged offsets the energy flowing back with a temperature gradient over the space between the two slabs. In our paper, the simulation box is firstly divided into N slabs ($N = 40$ in this paper) in the heat transfer direction, the first slab (slab 0) is defined as the "hot" slab, and the slab $N/2$ is defined as the "cold" slab. The temperature T in each slab is calculated by,

$$T = \frac{1}{3nk_B}\sum_{i=1}^{n} m_i v_i^2 \tag{2}$$

where the summation extends over the atoms i with masses m_i and velocities v_i. k_B is Boltzmann's constant. The heat flux is created by exchanging velocities of particles in "cold" and "hot" slabs. As what is introduced above, the "cold" slab donates its "hottest" particles with the highest kinetic energy to the "hot" slab in exchange for the latter's "coolest" particles with the lowest kinetic energy. This process leads to a temperature difference between the cool and the hot slab, and the temperature profile is given by the temperatures in the intervening slabs from 1 to $N/2$–1 and from $N/2 + 1$ to $N-1$, where the temperature is obtained by averaging temperatures within last 1000 simulation steps. With a time step of 0.2 fs, a total simulation time of 0.5 ns is taken to get a steady temperature distribution. To test whether the simulation system reaches the steady state or not, the effective thermal conductivity of the whole structure is also calculated and shown in Fig. 4. It shows that 0.5 ns are long enough to get a steady-state. After getting the temperature drop ΔT at the interface, the

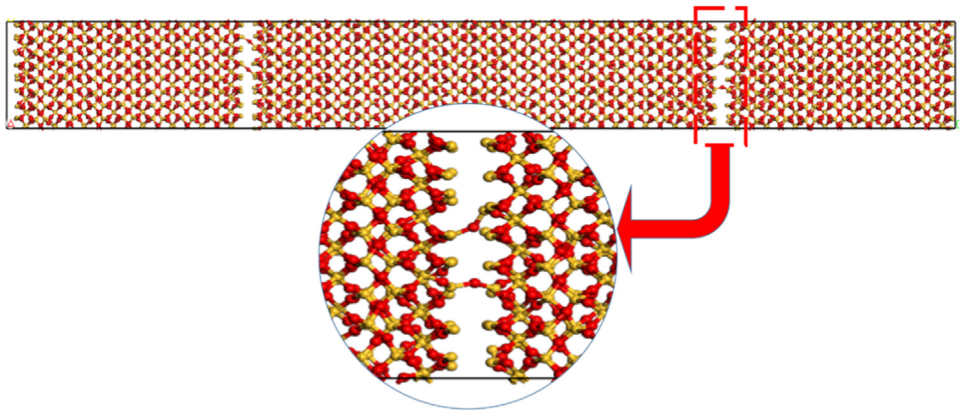

Fig. 2. Crystalline-crystalline silica configuration with a bonding ratio of 6.25%.

Table 2
The L-J parameters for Si and O atoms [18].

Atoms	ε (kcal/mol)	σ (Å)	$q(e)$
Si	0.08	4.2	1.1
O	0.04	3.6	−0.55

thermal contact resistance can be calculated by,

$$R = \frac{\Delta T}{q} \tag{3}$$

where ΔT is the temperature drop, q is the average heat flux within 1000 simulation steps. A large heat flux is applied here to clearly show the temperature jump at the interface, similar to that done in Ref. [15]. Extrapolating the linear temperature region to the interface, the temperature jump in crystalline-crystalline silica is shown in Fig. 3. All thermal contact resistance shown in this work is a mean value averaged over at least 5 separate simulations.

To further understand the thermal resistance at the interface, vibrational density of states (VDOS) is also calculated by using the Fourier transform of the velocity autocorrelation function (VACF),

$$\mathrm{VDOS}(\omega) = \int_{-\infty}^{+\infty} VACF(t)e^{-2\pi\omega t}dt \tag{4}$$

where t is time, ω is the frequency of phonons, VACF is obtained by,

$$\mathrm{VACF}(t) = \left\langle \frac{1}{N}\sum_{i=1}^{N} \vec{v_i}(t_0)\cdot\vec{v_i}(t_0+t) \right\rangle \tag{5}$$

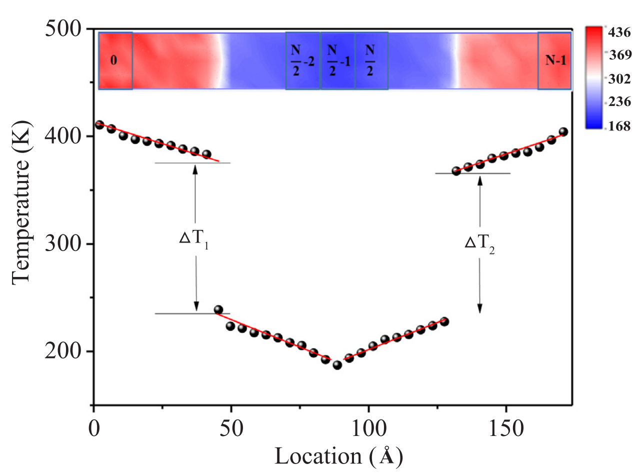

Fig. 3. Temperature distribution in crystalline-crystalline configuration at 300 K.

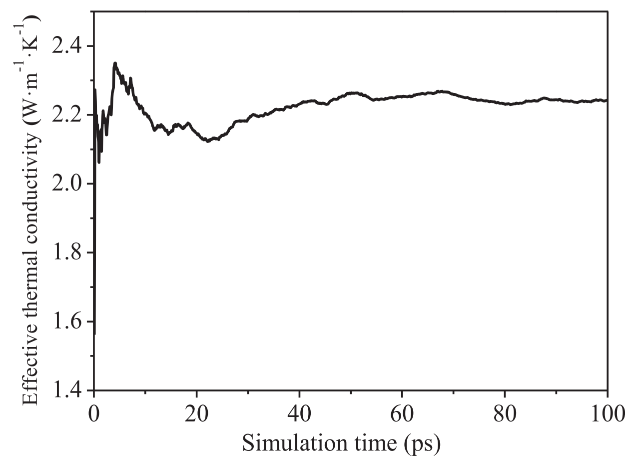

Fig. 4. Effective thermal conductivity of crystalline-crystalline silica.

where N is number of atoms, $\vec{v_i}(t_0)$represents the velocity of atom i at time t.

3. Results and discussions

In this part, firstly, the thermal contact resistance without considering chemical bonding at interface is investigated. Then, the effect of chemical bonds on thermal contact resistance is studied. Finally, the temperature dependence of thermal contact resistance is taken into account.

3.1. Thermal contact resistance without chemical bonding at interface

Without considering the chemical bonding at interface, thermal contact resistances of three different interface configurations as shown in Fig. 1 are listed in Table 3. It shows that the thermal contact resistance of crystalline-crystalline configuration is larger than that of other two configurations, and the amorphous-amorphous configuration

Table 3
Thermal contact resistance at 300 K for different configurations.

Configurations	Thermal contact resistance R ($\times 10^{-9}$ K m^2 W^{-1})
Amorphous-amorphous	5.61
Crystalline-amorphous	5.76
Crystalline-crystalline	6.06

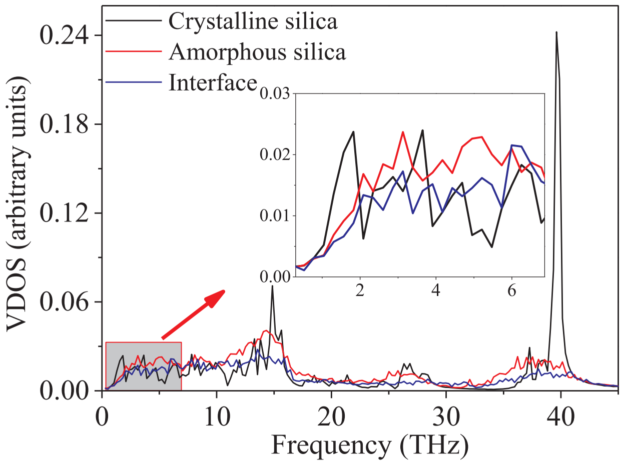

Fig. 5. Vibrational density of spectrum of crystalline silica, amorphous silica and interface.

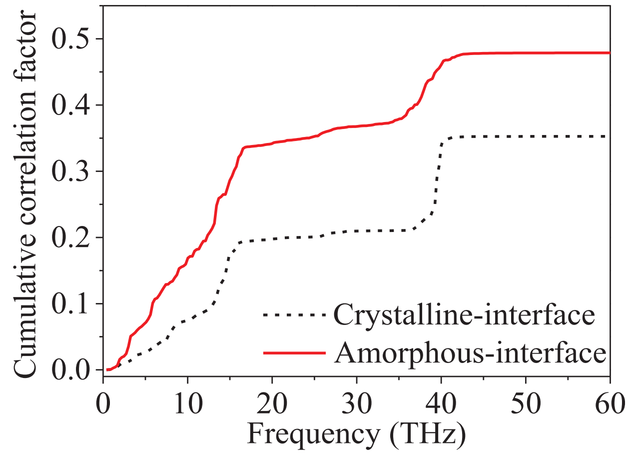

Fig. 6. Cumulative correlation factor between crystalline (amorphous) silica and interface.

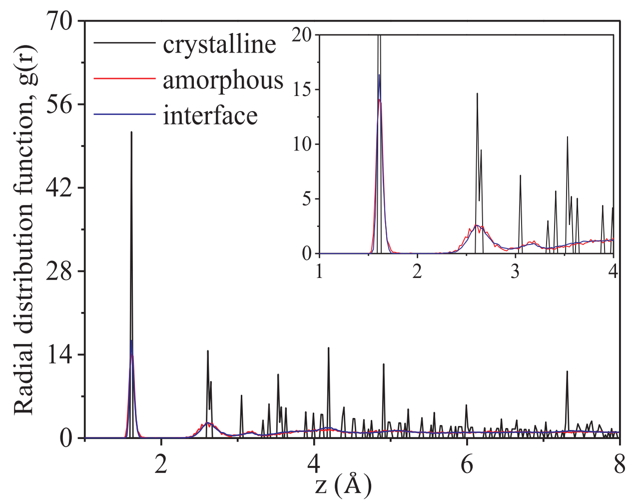

Fig. 7. Radial distribution function of crystalline silica, amorphous silica and interface.

has the lowest thermal contact resistance. It can be concluded that the same material with different morphologies (crystalline or amorphous) could lead to different thermal contact resistances, although this difference is less than 6%. Such a small difference of thermal contact resistance caused by different morphologies is negligible. Comparing thermophysical properties of different morphologies (crystalline or amorphous silica), it can be found that the difference of thermal contact resistance is much smaller than the difference of thermal conductivities, where the thermal conductivities of the crystalline and amorphous silica is 3.23 and 1.49 $\mathrm{W\,m^{-1}\,K^{-1}}$ respectively calculated by dividing heat flux with temperature gradient and sectional area in this paper. The calculated thermal conductivities of the crystalline and amorphous silica here are also very close to that got by MD simulations in Refs. [26,27] (3.5 $\mathrm{W\,m^{-1}\,K^{-1}}$ and 1.4 $\mathrm{W\,m^{-1}\,K^{-1}}$), which proves the applicability of the COMPASS potential for thermal transport simulation.

To reveal the mechanism of the thermal resistance, VDOS of the crystalline silica, the amorphous silica and the crystalline-amorphous interface are calculated and shown in Fig. 5. Here the interface VDOS is calculated from the velocity output of atoms located within the cutoff radius of L-J potential at interface (10 Å in this paper). In Fig. 5, there are more distinct crests and troughs in VDOS of crystalline silica than that of amorphous silica, and the interface VDOS is closer to that of amorphous silica than to crystalline silica, especially at the low frequency which dominants the thermal conductivity [28]. To estimate the VDOS match between interface and crystalline silica or amorphous silica, a cumulative correlation factor is calculated up to a specific cutoff frequency [29],

$$M(\varpi) = \frac{\int_0^{\varpi} VDOS_s(\omega)\cdot VDOS_i(\omega)d\omega}{\int_0^{\infty} VDOS_s(\omega)d\omega\cdot \int_0^{\infty} VDOS_i(\omega)d\omega} \quad (5)$$

where $VDOS_s$ represents the VDOS of crystalline or amorphous silica, $VDOS_i$ represents the VDOS of interface, ϖ represents a cutoff frequency. The result is shown in Fig. 6. It shows that for all frequencies, the VDOS of amorphous silica matches that of interface better. This good VDOS match leads to a lower thermal contact resistance of amorphous-amorphous configuration than crystalline-amorphous and crystalline-crystalline configurations, as shown in Table 3.

To illustrate the structure difference between crystalline or amorphous silica and interface, the RDF is calculated and shown in Fig. 7. Compared with the crystalline silica, the amorphous silica possesses a much closer RDF to that of interface (almost same). A closer RDF suggests that the interface structure is more similar to that of amorphous silica which should be responsible for the better VDOS match between amorphous silica and interface in Fig. 6.

3.2. Effect of chemical bonding

In this part, the influence of chemical bonding on thermal contact resistance is considered. Temperature profiles with different bonding ratios are shown in Fig. 8a–c, and the corresponding thermal contact resistances are displayed in Fig. 8d. Considering the mirror distribution of temperatures, only right side of distributions is show in Fig. 8a–c. Fig. 8a–c shows that the temperature drop is reduced by the increase of bonding ratios at interface, which means a decreased thermal contact resistance. In Fig. 8d, with the increase of the chemical bonding ratios, the thermal contact resistance firstly drops dramatically until arrives at a bonding ratio of about 20% where the thermal contact resistance is only about 15% that with a bonding ratio of 0%, and then gradually decreases to 0.

To reveal the effect of chemical bonding, the SED of crystalline-crystalline interface is calculated. Here the interface SED is calculated from the velocity output of atoms located within the cutoff radius of L-J potential at interface (10 Å in this paper). The frequency resolution is selected to be 0.01 THz with 0.1 ns NVE simulations, and the wavenumber is $0.02\kappa/(2\pi/a)$, where $\kappa = 2\pi n/aN$, a is the unit cell constant selected to be cutoff radius of L-J potential (10 Å), N is the total number of unit cells along the heat transfer direction (N = 50 in this work), n is

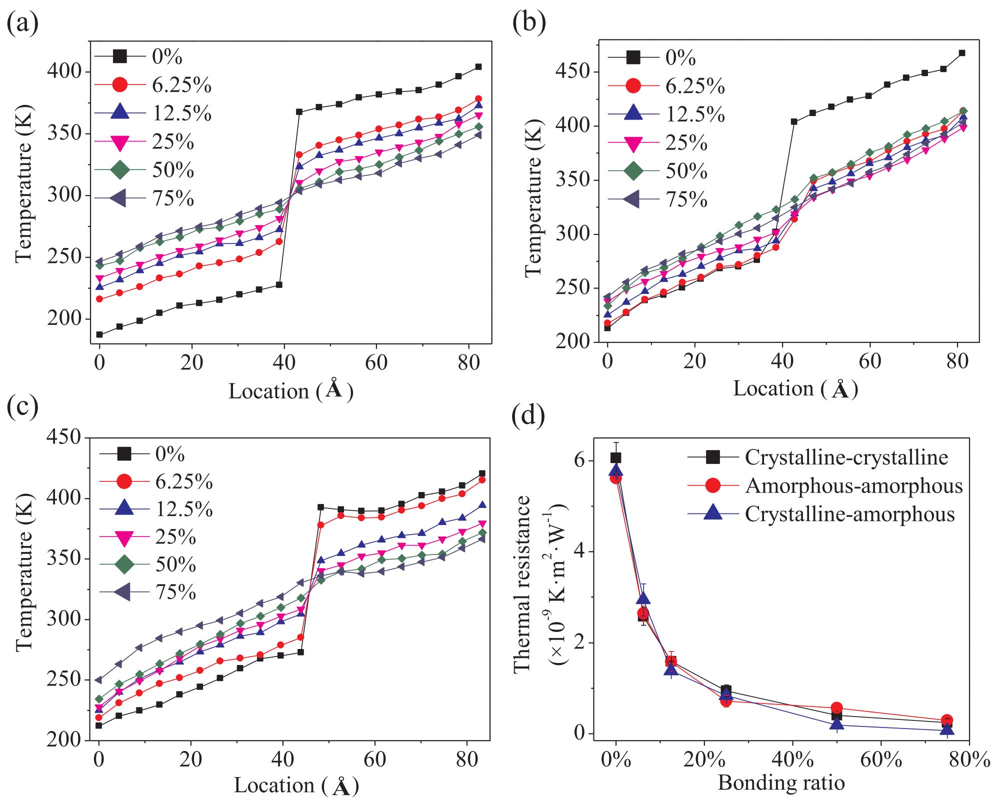

Fig. 8. Thermal contact resistance with different bonding ratios. (a) temperature profiles of crystalline-crystalline configurations; (b) temperature profiles of amorphous-amorphous configurations; (c) temperature profiles of crystalline-amorphous configurations; (d) thermal contact resistance.

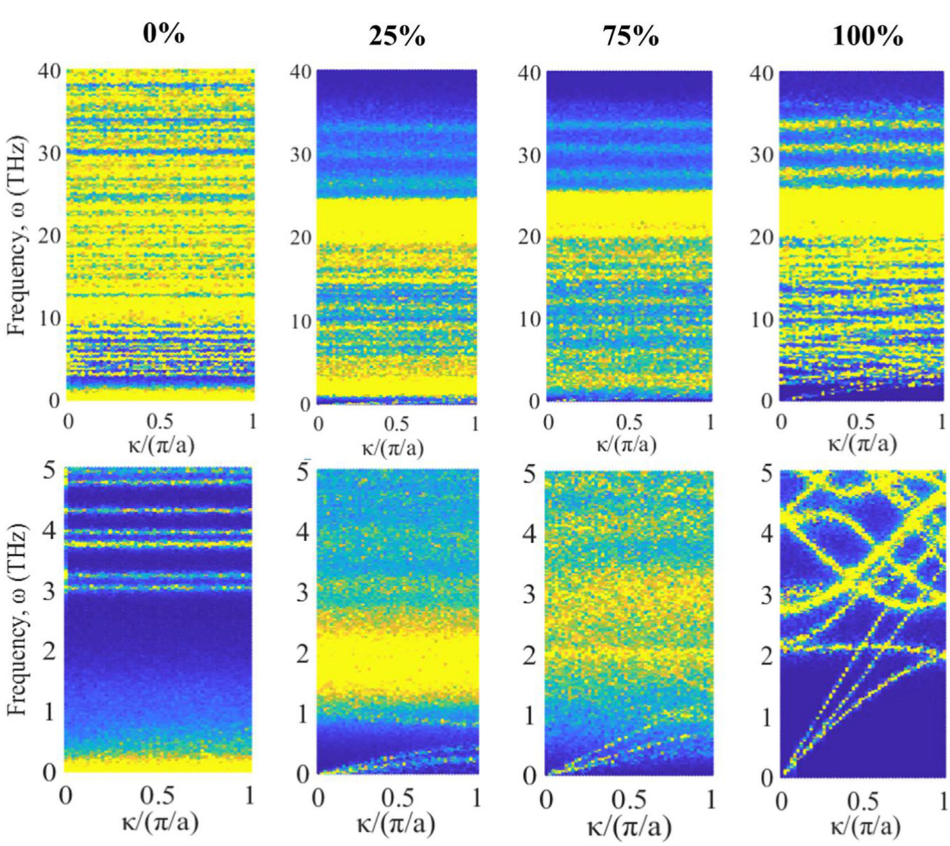

Fig. 9. Interfacial SEDs with different chemical bonding ratios (only frequencies less than 40 THz are shown here).

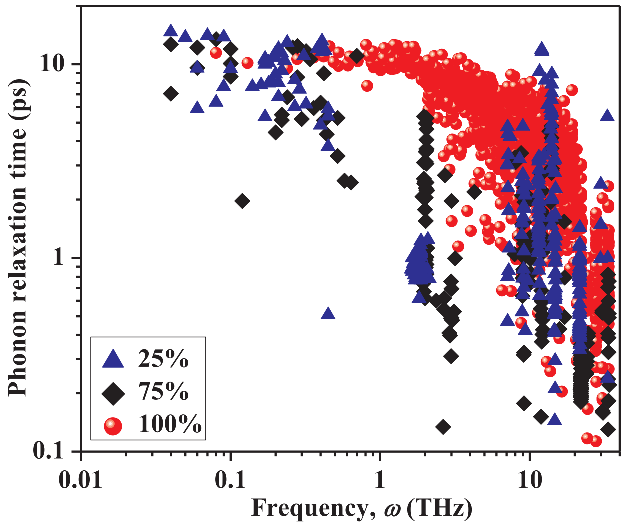

Fig. 10. Phonon relaxation time with different bonding ratios.

an integer ranging from 0 to N − 1. More details about the calculation can be found in Refs. [30–33]. The SED with bonding ratios of 0, 25, 75 and 100% are shown in Fig. 9. The structure with a bonding ratio of 100% is actually a material with no interface. Thus, there are 3 acoustic branches for a bonding ratio of 100% in Fig. 9. Considering the heat is mainly carried by acoustic phonons, only acoustic branches are analyzed here. For a bonding ratio of 0%, no chemical bonds are established between two materials, and therefore there are no obvious acoustics branches in Fig. 9. With the increase of bonding ratios, the frequency span of acoustic branches increases from less than 0.5 THz (25%) to 4 THz (100%). This suggests an increase of phonon velocity which can be approximately thought as the slope of the branch in Fig. 9. As we know, based on the kinetic theory, the thermal transport depends on the phonon velocity and phonon relaxation time. To reveal the influence of chemical bonding on phonon relaxation time, the shape of peak and valley profiles of SED at all branches is fitted to the Lorentzian function to obtain the phonon relaxation time. Results are shown in Fig. 10, the fitting method can be found in Refs. [30–32]. In the frequency less than 1 THz, there is no much difference of phonon relaxation time between different bonding ratios. From the analyzation of phonon velocity and relaxation time at interface, it can be concluded that the increase of phonon velocity should be responsible for the decrease of thermal contact resistance.

3.3. Influence of temperature

In this part, we consider the influence of temperature on the thermal contact resistance. The thermal contact resistances for bonding ratios of 0% and 25% at different temperatures are depicted in Fig. 11. It shows that the thermal contact resistance decreases with increasing temperature. Similar phenomena were also observed in Refs. [34,35] for bonding ratio of 0%. With the temperature increasing from room temperature to a higher value, although the mean free path of phonons becomes short because of the severe phonon-phonon scattering, the thermal conductance at interface still rises due to the increase of high-frequency phonons which possesses a higher transmissivity. This high transmissivity can be explained by the good VDOS match at high frequency between crystalline silica or amorphous silica and interface.

4. Conclusion

In this paper, we employ a non-equilibrium molecular dynamics (NEMD) method to study the effect of chemical bonding on thermal contact resistance. Results show that the chemical bonding can greatly affect the thermal contact resistance. With the increase of the chemical bonding ratios, the thermal contact resistance firstly drops dramatically until arrives at a bonding ratio of about 20% where the thermal contact resistance is only about 15% that with a bonding ratio of 0%, and then gradually decreases to 0. Analyzing the spectral energy density of phonons at the interface, we conclude that the increase of the phonon velocity should be responsible for the decrease of the thermal contact resistance.

With a bonding ratio of 0%, the thermal contact resistance of crystalline-crystalline configuration is larger than that of other two configurations (amorphous-amorphous and crystalline-amorphous), and the amorphous-amorphous configuration has the lowest thermal contact resistance. Although the same material with different morphologies (crystalline or amorphous) could lead to different thermal contact resistances, this difference is less than 6% small enough to be neglected. Such a small difference of thermal contact resistances between different morphologies is due to the different VDOS match.

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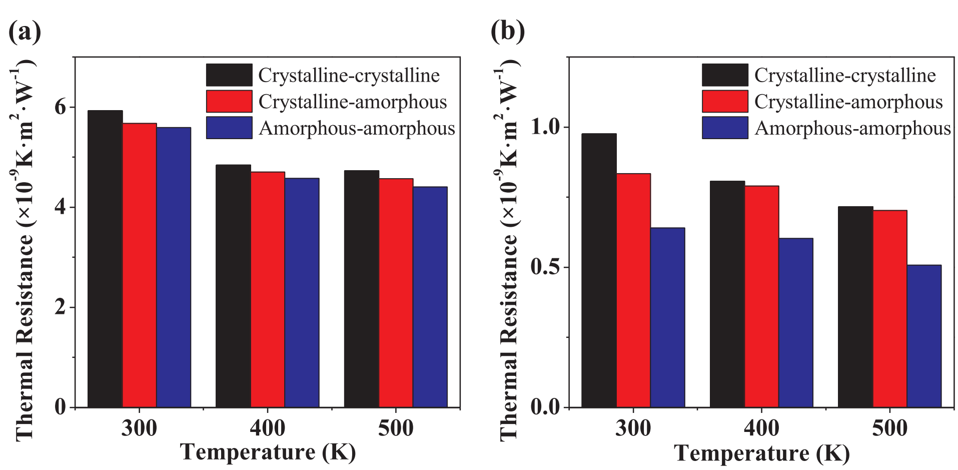

Fig. 11. Interfacial thermal contact resistances at different temperatures with bonding ratios of (a) 0% and (b) 25%.

Appendix A. The interfacial bonding ratio

The interfacial bonding ratio is defined as the percentage of atoms connected through Si—O—Si bonds to other-side atoms at interface. As shown in Fig. A1, the silicon atom which is connected with 4 atoms at the interface means a chemical bonding for the atom, while connecting with 3 atoms means a non-bonding. Thus, the bonding ratio could be calculated by,

$$\text{Bonding ratio} = 1-\frac{\text{number of silicon atoms which are connected with 3 oxygen atoms}}{\text{total number of silicon atoms at the interface}} \tag{A1}$$

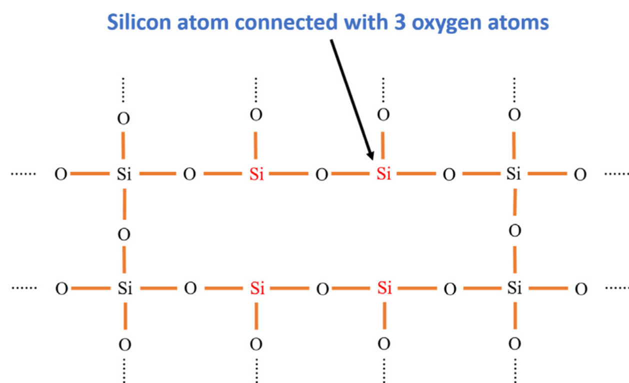

Fig. A1. Schematic of the interface.

Appendix B. The COMPASS potential

The COMPASS force field used in this work could be expressed as,

$$E_{total} = \sum_{b}[k_{b2}(b-b_0)^2 + k_{b3}(b-b_0)^3 + k_{b4}(b-b_0)^4] + \sum_{\theta}[k_{\theta2}(\theta-\theta_0)^2 + k_{\theta3}(\theta-\theta_0)^3 + k_{\theta4}(\theta-\theta_0)^4] + \sum_{\phi}[k_{\phi1}(1-\cos\phi) + k_{\phi2}(1-\cos2\phi) + k_{\phi3}(1-\cos3\phi)] + \sum_{\chi}k_{\chi2}\chi^2 + \sum_{b,b'}k_{b,b'}(b-b_0)(b'-b_0) + \sum_{b,\theta}k_{b,\theta}(b-b_0)(\theta-\theta_0) + \sum_{b,\phi}(b-b_0)[k_{b,\phi1}\cos\phi + k_{b,\phi2}\cos2\phi + k_{b,\phi3}\cos3\phi] + \sum_{\theta,\phi}(\theta-\theta_0)[k_{\theta,\phi1}\cos\phi + k_{\theta,\phi2}\cos2\phi + k_{\theta,\phi3}\cos3\phi] + \sum_{b,\theta}k_{b,\theta}(\theta'-\theta_0) + \sum_{\theta,\theta,\phi}k_{\theta,\theta,\phi}(\theta-\theta_0)(\theta'-\theta_0)\cos\phi + \sum_{i,j}\varepsilon_{ij}\left[2\left(\frac{\sigma_{ij}}{r_{ij}}\right)^9-3\left(\frac{\sigma_{ij}}{r_{ij}}\right)^6\right] + \frac{1}{4\pi\varepsilon_0}\sum_{i,j}\frac{q_iq_j}{r_{ij}} \tag{B1}$$

where the subscripts represent the bond (b), angle (θ), dihedral angle (ϕ) and out-of-plane angle (χ). The last two terms in Equation B1 are non-bonding terms. The corresponding parameters in Equation B1 are listed in Table B1.

Table B1
Parameters for COMPASS force field with international system of units.

Parameter	Value	Parameter	Value	Parameter	Value	Parameter	value
b_0	1.6562	k_{b2}	114.21.64	k_{b3}	140.4212	k_{b4}	80.2685
θ_0	114.2676	$k_{\theta2}$	2.9501	$k_{\theta3}$	−15.5949	$k_{\theta4}$	0
$k_{\phi1}$	−0.225	$k_{\phi2}$	0	$k_{\phi3}$	−0.01	$k_{\chi2}$	0
$k_{b,b'}$	0	$k_{b,\theta}$	28.6686	$k_{b,\phi1}$	0	$k_{b,\phi2}$	0
$k_{b,\phi3}$	−0.6302	$k_{\theta,\phi1}$	0	$k_{\theta,\phi2}$	0	$k_{\theta,\phi3}$	0.3382
$k_{\theta,\theta,\phi}$	−14.0484						

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